

MT 505: Thin Film Technology

Course Code	MT 505
Course Name	Thin Film Technology
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (Elective) /M.Tech./Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** This course provides a comprehensive understanding of the science and technology behind the growth of inorganic solid thin films and coatings. It introduces key characterization techniques for real-time and post-growth analysis. The second part emphasizes project-based learning, exploring thin film applications tailored to student interests. Key concepts in vacuum technology, components, and plasma will also be introduced, as these are essential for understanding thin film deposition processes. [6 hours]
- **Thin film growth:** Introduction of thin film deposition techniques: Physical Vapor Deposition (PVD) methods like sputtering and ion-beam, and Chemical Vapor Deposition (CVD) processes such as APCVD, MOCVD, and PECVD. Solution-based methods include ink-jet printing, spin coating, and spray pyrolysis. Other advanced techniques include Atomic Layer Deposition (ALD), Pulsed Laser Deposition (PLD), and Molecular Beam Epitaxy (MBE). Along with these techniques, it is essential to understand the basic mechanisms of thin film growth from the vapour phase for a complete understanding of the process. [12 hours]
- **Thin film characteristics:** Introduction of optical, physio-chemical, mechanical, and electrical properties of thin films. Optical properties: optical microscope, and UV-Vis Spectroscopy for absorption, transparency, and refractive index. Physio-chemical properties: thickness, film density, chemical composition, structure, and surface morphology, analyzed through Ellipsometry, XRR, XPS, XRD, SEM, TEM, and AFM. Electrical properties involve conduction phenomena and Hall measurements. Mechanical properties address thin film stress and wafer curvature using Stoney's equation, bulge tests, and other methods to understand residual stresses in thin films. [12 hours]
- **Thin film applications:** Thin films are widely used in batteries, resistors, solar cells, and photovoltaics. They also play a key role in Microelectromechanical (MEMS) and Nanoelectromechanical (NEMS) systems, sensors, actuators, memristors, diodes, thin film transistors, and display technologies, contributing to advancements in electronics, energy, and display industries. [8 hours]

Laboratory Modules

1. NA

Text and Reference Books

1. **Thin Film Materials: Stress, Defect Formation and Surface Evolution**
2. Fredrick Madaraka Mwema, Tien-Chien Jen, Lin Zhu, **Thin Film Coatings: Properties, Deposition, and Applications**, (Emerging Materials and Technologies), CRC Press, 2022.
3. Krishna Seshan, Dominic Schepis, **Handbook of Thin Film Deposition**, William Andrew, 2018.
4. K. S. Sree Harsha, **Principles of Vapor Deposition of Thin Films**, Elsevier, 2006.
5. K. Seshan, **Handbook of Thin-Film Deposition Processes and Techniques**, William Andrew Publishing, 2002.
6. John E. Mohan, **Physical Vapor Deposition of Thin Film**, John Wiley & Sons, 2000

Any other Study Material

1. Students will be supplied with published journal articles as reference materials to support their studies.